A Mechanistic Model of Molecular Crowding in the DNA Predator and Prey Oscillator

How crowding reshapes the Fujii and Rondelez PEN oscillator: the model we propose, and how it was built.

BIOMOD dry lab. Written for a project teammate. Implementation: crowded_mechanistic.py

1. What this document is

This describes the model our team built for the dry lab part of the project: a model of how molecular crowding (adding an inert polymer such as PEG or Ficoll to the tube) changes the DNA predator and prey oscillator. It is written for a teammate who knows the biology at a high level but has not seen how this model was put together. By the end you will know exactly which equations we propose, what every symbol means, and why each piece is there.

The structure is: Section 2 states the final model in full. Sections 3 to 8 derive it, starting from the published dilute oscillator and adding the crowding physics one layer at a time. Sections 9 to 11 give the predictions, the optional simpler version, and what the wet lab should measure to test it.

2. THE FINAL MODEL (THE EQUATIONS WE PROPOSE)

This is the complete model. If you read only one section, read this one. Everything after it explains where each piece comes from.

The model tracks seven concentrations in nanomolar (nM): free prey strand N, free predator strand P, free template G, and four intermediate complexes (two molecules bound together): C_{NG} (prey on template), C_{NP} (prey on predator), C_{NE} (prey on the degrading enzyme), and C_{PE} (predator on the degrading enzyme). The free enzyme is whatever is not currently bound:

$$E = rec - C_{NE} - C_{PE}$$

The seven concentrations change in time according to these equations. Each k is a rate constant (defined just below); the crowding-sensitive ones are highlighted in the next box.

$$\frac{dN}{dt} = -k_{on}^{NG} NG + k_{off}^{NG} C_{NG} + 2k_{cat}^{g} C_{NG} - k_{on}^{NP} NP + k_{off}^{NP} C_{NP} - k_{on}^{eN} NE + k_{off}^{eN} C_{NE} + \lambda_N$$

$$\frac{dP}{dt} = -k_{on}^{NP} NP + k_{off}^{NP} C_{NP} + 2k_{cat}^{p} C_{NP} - k_{on}^{eP} PE + k_{off}^{eP} C_{PE} + \lambda_P$$

$$\frac{dG}{dt} = -k_{on}^{NG} NG + k_{off}^{NG} C_{NG} + k_{cat}^{g} C_{NG}$$

$$\frac{dC_{NG}}{dt} = k_{on}^{NG} NG - k_{off}^{NG} C_{NG} - k_{cat}^{g} C_{NG}$$

$$\frac{dC_{NP}}{dt} = k_{on}^{NP} NP - k_{off}^{NP} C_{NP} - k_{cat}^{p} C_{NP}$$

$$\frac{dC_{NE}}{dt} = k_{on}^{eN} NE - k_{off}^{eN} C_{NE} - k_{cat}^{eN} C_{NE}$$

$$\frac{dC_{PE}}{dt} = k_{on}^{eP} PE - k_{off}^{eP} C_{PE} - k_{cat}^{eP} C_{PE}$$

2.1 What each symbol means

- **N, P:** free prey and predator strands. These are the two populations we plot.
- **G:** the free template strand that prey is copied on.
- **E:** the free exonuclease (ttRecJ), the single enzyme that degrades both prey and predator.
- **C_{NG}, C_{NP}, C_{NE}, C_{PE}:** a complex, meaning two molecules currently bound together. For example C_{NG} is one prey strand stuck to the template, and C_{PE} is a predator strand held by the enzyme.
- **k_{on}, k_{off}, k_{cat}:** the on-rate (how fast two molecules bind), the off-rate (how fast a complex falls back apart), and the catalytic rate (how fast a bound complex does its chemistry). The superscript names which pair it belongs to.
- **lambda_N, lambda_P:** a small constant background synthesis of prey and predator (a 'leak'). It is tiny next to the roughly 130 nM swings but it keeps the rhythm from collapsing (Section 8).
- **rec, G_{tot}:** the total exonuclease and total template put into the tube (fixed numbers, not variables).

2.2 What each term in the prey and predator equations means

Take the prey equation dN/dt. Its terms come in groups, one group per thing prey can do:

- **With template (bind/unbind):** prey binds the template and leaves the free pool ($-k_{on}^{NG}NG$), or that complex falls back apart and returns the prey ($+k_{off}^{NG}C_{NG}$).
- **Growth ($+2k_{cat}^{g}C_{NG}$):** the polymerase and nickase copy the bound prey and release the product as two strands. One prey went in, two come out, so this is the net growth of prey.
- **With predator:** prey binds and unbinds the predator, because the predator uses prey as its template ($-k_{on}^{NP}NP + k_{off}^{NP}C_{NP}$). Prey is consumed when this complex goes on to make predator (in the predator equation).
- **With the enzyme:** prey binds and unbinds the exonuclease before being cut ($-k_{on}^{eN}NE + k_{off}^{eN}C_{NE}$).
- **Leak:** the small constant background (+lambda_N).

The predator equation dP/dt reads the same way: it binds and unbinds prey and replicates ($+2k_{cat}^{p}C_{NP}$, each event making two predator strands), binds and unbinds the enzyme ($-k_{on}^{eP}PE + k_{off}^{eP}C_{PE}$), and has its own leak (+lambda_P).

2.3 What the complex equations mean

Each complex is made when its two partners bind, and is removed in two ways: it can simply fall apart, or it can react. So every complex follows the same shape, made minus (unbind plus react). For the prey on template:

$$\frac{dC_{NG}}{dt} = k_{on}^{NG} NG - \left(k_{off}^{NG} + k_{cat}^{g}\right) C_{NG}$$

and the other three complexes are identical in form. The free enzyme is not its own equation; it is fixed at every instant by conservation, E = rec - C_{NE} - C_{PE}, which is what makes the enzyme a shared, limited resource and drives the oscillation.

How crowding enters: one number, the crowder volume fraction phi

Crowding is summarised by the volume fraction phi that the crowder occupies, set by its weight concentration C and specific volume v-bar:

$$\varphi = \overline{v}\frac{C}{100}$$

Every rate constant above is then the dilute value times physics that depends only on phi, the crowder size, and the molecules' sizes. For each binding step, the binding strength (equilibrium constant K) and the on-rate change as:

$$K(\varphi) = K^0 \exp\left(\quad -\frac{\Delta\Delta G_{EV}}{RT}\right), k_{on}(\varphi) = \frac{k_{on}^0}{\eta_{eff}}, k_{off} = \frac{k_{on}}{K}$$

where the excluded-volume free energy that strengthens binding is

$$\frac{\Delta\Delta G_{EV}}{RT} = \mu_{ex}(r_{AB}) - \mu_{ex}(r_A) - \mu_{ex}(r_B)$$

and the catalytic constants shift by enzyme class (polymerase and nickase up, exonuclease down):

$$k_{cat}^{g}, k_{cat}^{p} \rightarrow k_{cat} e^{+c_P \varphi}, k_{cat}^{eN}, k_{cat}^{eP} \rightarrow k_{cat} e^{-c_E \varphi}$$

Reading each crowding rule:

- **phi (volume fraction):** the fraction of the volume taken up by crowder. It is the one knob the whole study turns. v-bar is the crowder's specific volume and C is its weight percent.
- **K(phi):** the binding strength. K^0 is its dilute value; the exponential makes binding stronger as crowding squeezes the two partners together (Section 6.1).
- **k_{on}(phi):** the on-rate, slowed by dividing by the effective viscosity eta_eff: thicker solution, slower meeting (Section 6.2). The off-rate k_{off} is then set to k_{on}/K so the kinetics and the binding strength stay consistent.
- **Delta-Delta-G_{EV}:** how much the binding strength changes, equal to the extra room the compact complex gives back to the crowders (Section 6.1).
- **k_{cat} factors:** the catalytic steps. Crowding speeds the polymerase and nickase up (factor e^{+c_{P} phi}) and slows the exonuclease down (factor e^{-c_{E} phi}). c_P and c_E are how strong each effect is, taken from enzyme measurements.

mu_ex and eta_eff are defined in Section 6. At zero crowder (phi = 0) every rate constant returns to its dilute value (Table in Section 5) and the model reproduces the measured oscillation: period about 90 minutes, predator swinging from about 22 up to 154 nM. That is the whole model.

3. Background: the oscillator we start from

The system (Fujii and Rondelez, 2013) is three short DNA species and three enzymes in a buffer. Prey N copies itself on a template G with the help of a polymerase and a nicking enzyme. Predator P copies itself using prey N as its template, consuming prey in the process. A single exonuclease (ttRecJ) degrades both prey and predator. Because the predator lags the prey, the two populations chase each other up and down in a sustained oscillation, exactly like foxes and rabbits.

The published description, after simplification, is two equations for the prey and predator concentrations. We use these as our dilute baseline:

$$\frac{dN}{dt} = \frac{k_1 polGN}{1 + bGN} - k_2 polNP - \frac{reck_N N}{1 + P/K_{m,P}}$$

$$\frac{dP}{dt} = k_2 polNP - \frac{reck_P P}{1 + P/K_{m,P}}$$

Here pol is the polymerase level and rec is the exonuclease level. The six constants k_1, b, k_2, k_N, k_P, $K_{m,P}$ are 'lumped': each one secretly bundles several physical steps together. That bundling is the key point for what follows.

4. The approach: act on the chemistry, not on the published constants

The tempting shortcut is to keep the two equations above and just multiply each lumped constant by some crowding factor. We deliberately do not do that, for a concrete reason: crowding does not act on a lumped constant. It acts on the separate physical steps hidden inside it, and those steps respond to crowding in opposite directions.

Take k_1 (prey growth). It hides a binding step (prey finding its template), a catalytic step (the polymerase copying it), and a diffusion step (the two molecules meeting). Crowding speeds binding up (by excluded volume) but slows diffusion down (by viscosity), and separately changes the enzyme's turnover. A single fitted factor on k_1 cannot represent three effects pulling different ways.

So instead we expand the lumped model back into the elementary reactions it came from (Section 5), and let crowding act on each elementary step through shared physics (Section 6). The payoff: the same physics that stabilises a DNA duplex also tightens enzyme binding, and the same viscosity that slows the polymerase slows the exonuclease. Nothing is fit step by step; it all follows from phi, the crowder size, and the molecules' sizes. This is also what lets two crowders of the same volume fraction (PEG versus Ficoll) give genuinely different answers.

5. The reaction network

These are the elementary reactions. A double arrow means reversible binding (on-rate forward, off-rate back); a single arrow is the irreversible catalytic step. Polymerase and

nickase are plentiful here, so they are folded into the catalytic constants rather than tracked separately.

Prey growth: $N + G \rightleftharpoons C_{NG} \rightarrow G + 2\,N$ (on k_{on}^{NG}, off k_{off}^{NG}, copy k_{cat}^{g})
Predation: $N + P \rightleftharpoons C_{NP} \rightarrow 2\,P$ (on k_{on}^{NP}, off k_{off}^{NP}, copy k_{cat}^{p})
Prey decay: $N + E \rightleftharpoons C_{NE} \rightarrow E$ (on k_{on}^{eN}, off k_{off}^{eN}, cut k_{cat}^{eN})
Predator decay: $P + E \rightleftharpoons C_{PE} \rightarrow E$ (on k_{on}^{eP}, off k_{off}^{eP}, cut k_{cat}^{eP})
Conserved totals: $G_{tot} = G + C_{NG}$, $rec = E + C_{NE} + C_{PE}$

Two features of the published model now appear on their own, instead of being written in by hand:

- **Template saturation.** the saturating growth term $1/(1 + b\,G\,N)$ is simply finite template: once the template G_{tot} is fully occupied, growth cannot speed up.
- **The oscillation engine.** the competition term $1/(1 + P/K_{m,P})$ is the shared exonuclease being a limited resource. The predator binds it tightly (high affinity) and the prey binds it weakly. When predator is abundant it occupies the enzyme and shields the prey from being degraded. That self-shielding is the engine of the oscillation, and here it falls out of conservation with no extra assumption.

Mathematically, applying a quasi-steady-state assumption to the four complexes turns these reactions back into the two-variable equations of Section 3, which fixes the dilute values of every elementary constant. Those dilute values are:

Lumped constant	Built from elementary steps	Dilute value
k_1 (prey growth)	k_{cat}^{g} times K_1 (K_1 = N,G binding)	2.0e-3
b (growth saturation)	K_1 / G_{tot}	4.8e-5
k_2 (predation)	k_{cat}^{P} times K_2 (K_2 = N,P binding)	3.1e-3
k_N (prey decay)	k_{cat}^{eN} / K_m^{N} (weak binding, K_m^{N} large)	2.1e-2
k_P (predator decay)	k_{cat}^{eP} / K_m^{P}	4.7e-3
$K_{m,P}$ (competition)	$(k_{off}^{eP} + k_{cat}^{eP}) / k_{on}^{eP}$	34 nM

6. The crowding physics (where the crowding rules come from)

6.1 Excluded volume: why crowding strengthens binding

A crowder molecule physically occupies space, so a reacting molecule has less room. The free-energy cost of inserting a molecule of radius r into a bath of crowders (radius R_c) packed at fraction phi is given by scaled-particle theory (Minton; Lebowitz):

$$\frac{\mu_{ex}(r)}{RT} = -\ln(1-\varphi) + (3z+3z^2+z^3)\frac{\varphi}{1-\varphi}, z = \frac{r}{R_c}$$

When two molecules A and B bind into a more compact product AB, the product takes up less room than the two separately, so binding 'gives back' space to the crowders. That is favourable, so crowding pushes binding forward. The size of that push is the difference in insertion cost (the equation used in Section 2):

$$\frac{\Delta\Delta G_{EV}}{RT} = \mu_{ex}(r_{AB}) - \mu_{ex}(r_A) - \mu_{ex}(r_B)$$

This one function supplies the excluded-volume effect for every binding step in the network, strand pairing and enzyme binding alike, differing only by the molecules' radii. The crowder radius R_c enters through $z = r/R_c$, which is exactly why a small crowder (PEG) and a large one (Ficoll) at the same phi give different results: smaller R_c means a larger excluded-volume push per unit phi.

6.2 Viscosity: why crowding slows the on-rates

Crowding also makes the solution thicker, so molecules diffuse and meet more slowly. For a polymer mesh (PEG) the effective viscosity felt by a molecule depends on its size relative to the mesh spacing xi (Holyst):

$$\frac{\eta_{eff}(r)}{\eta_0} = \exp\left[b\left(\frac{R_{eff}}{\xi}\right)^a\right], \frac{1}{R_{eff}^2} = \frac{1}{R_h^2} + \frac{1}{r^2}, \xi = R_c\left(\frac{c}{c^*}\right)^{-0.75}$$

A binding on-rate then slows in proportion to 1/eta_eff at the size of the larger partner (the slower mover sets the meeting rate). Small strands barely notice; large enzymes feel it strongly. A compact sphere like Ficoll has no mesh, so it uses the hard-sphere (Mooney) law instead:

$$\frac{\eta}{\eta_0} = \exp\left[\frac{[\eta]\varphi}{1 - \varphi/\varphi_{max}}\right]$$

6.3 Smaller effects (kept secondary)

Crowding also lowers the activity of water and can sequester buffer ions, both of which nudge DNA pairing. For the short strands here these are modest corrections, so they are included but kept small; the ion channel is switched off by default because our buffer uses magnesium, whereas the available data are for sodium.

7. Putting it together: how crowding moves each rate constant

Each elementary constant now changes with phi in a fixed way. For binding steps, the strength changes by the excluded-volume free energy, the on-rate slows by viscosity, and the off-rate is set so the two stay consistent:

$$K(\varphi) = K^0\exp\left(-\frac{\Delta\Delta G_{EV}}{RT}\right), k_{on}(\varphi) = \frac{k_{on}^0}{\eta_{eff}}, k_{off} = \frac{k_{on}}{K}$$

The catalytic step is intrinsic enzyme chemistry, which excluded volume cannot predict, so it is taken from enzyme measurements: crowding raises polymerase and nickase turnover and lowers exonuclease turnover.

$$k_{cat}^{pol/nick}(\varphi) = k_{cat}e^{+c_P\varphi}, k_{cat}^{exo}(\varphi) = k_{cat}e^{-c_E\varphi}$$

The net effect on the lumped constants, which the code reads off automatically, is that for PEG the growth and predation constants (k_1, k_2) rise while the decay constants (k_N, k_P) fall,

because the exonuclease is being slowed. The slowing of the exonuclease is the single most important consequence, since the oscillation lives or dies on it.

8. Calibration: two small adjustments

Two adjustments turn the elementary constants into a model whose dilute behaviour matches the experiment.

- **Timescale (0.8).** a single factor of 0.8 multiplies the catalytic constants. The quick estimate of those constants ignores the prey and predator that sit temporarily inside the complexes; tracking that bound pool slightly slows the cycle, and this factor restores the measured period.
- **Leak.** a tiny constant background synthesis (lambda_N about 0.02, lambda_P about 0.006 nM per minute), a documented feature of this enzyme toolbox, lifts the populations off zero. Without it the predator can fall to exactly zero and never recover; with it the oscillation is robust, as seen in experiment.

Result at zero crowder: period 90 minutes, predator swinging from about 22 up to 154 nM, a strong steady oscillation, matching the published behaviour.

9. Predictions

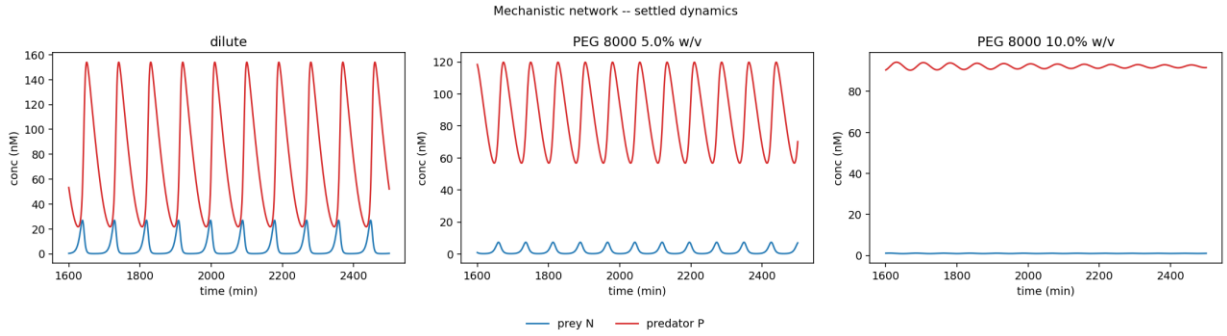


Figure 1. Settled dynamics at zero crowder and at 5 percent and 10 percent PEG 8000. The swings shrink as crowding increases.

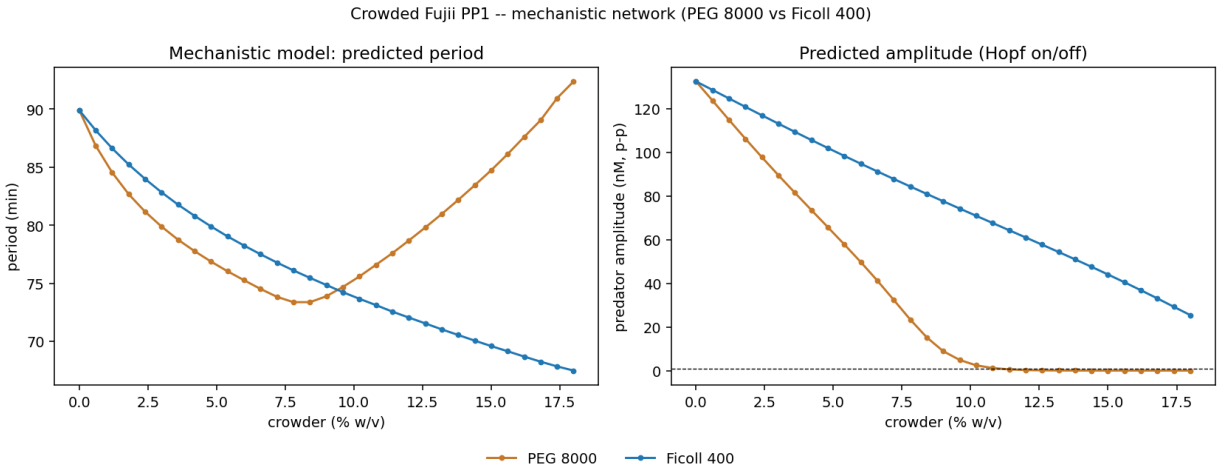


Figure 2. Period (left) and predator wave size (right) versus crowder concentration. PEG 8000 drives the wave size to zero near 10 to 12 percent (the oscillation stops); Ficoll 400 keeps oscillating throughout.

Condition	Period (min)	Predator wave (nM)	Oscillating?
Dilute (no crowder)	90	132	yes, strong
5 percent PEG 8000	78	76	yes
8 percent PEG 8000	73	19	yes, near the edge
About 11 percent PEG 8000	stops	about 0	no, waves switch off
16 percent Ficoll 400	69	38	yes, still going

The three testable claims for the wet lab: (1) crowding shortens the period; (2) PEG 8000 switches the oscillation off near 10 to 12 percent; (3) at the same volume fraction, Ficoll 400 is far milder and keeps the oscillation alive. The third claim, the same packing giving opposite outcomes for two crowders, is the novel and falsifiable result.

10. If you want a simpler version

If the full seven-equation model is more than an analysis needs, it can be collapsed to the two-variable form of Section 3 by assuming binding is fast, with the crowding-dependent constants given by the rules above, for example:

$$k_1(\varphi) = \frac{k_{cat}^{g}(\varphi)K_1(\varphi)}{pol}, b(\varphi) = \frac{K_1(\varphi)}{G_{tot}}, k_2(\varphi) = \frac{k_{cat}^{p}(\varphi)K_2(\varphi)}{pol}$$

Note: Important. This two-variable shortcut is good for intuition and for the form of each constant's crowding dependence, but it is not quantitatively faithful here. On its own it mispredicts even the dilute period (about 45 instead of 90 minutes) and loses the oscillation almost immediately, because it cannot hold the prey and predator that are temporarily bound inside the complexes. All quantitative predictions (period, the switch-off concentration, the PEG versus Ficoll difference) come from the full seven-species model. The interactive explorer runs that full model.

11. Assumptions and what to measure

- **Enzymes.** polymerase and nickase are plentiful and folded into the catalytic constants; only the exonuclease is treated as a limited, saturating resource.
- **Intermediates.** fast intermediate folding states are not tracked (assumed quick).

- **Enzyme data.** the strength of the exonuclease slowdown is taken from a related enzyme (Exonuclease I), not ttRecJ itself.
- **Tunable inputs.** the excluded-volume radii, the viscosity constants, the catalytic slopes, and the leak size are all literature-anchored but tunable. The qualitative story (period shortens, PEG switches off, Ficoll milder) is robust to their exact values; the precise switch-off concentration is not, so it should be fitted.
- **Known gaps.** all Ficoll 400 numbers are extrapolations (no direct Ficoll 400 data in the surveyed papers).

Highest-value measurements, in order: (1) ttRecJ activity versus PEG 8000 and versus Ficoll 400; (2) prey and predator pairing strength versus crowder; (3) solution viscosity versus crowder at the reaction temperature; (4) the background leak rate.

12. Key references

Access tags: [full] full text seen, [abs] abstract only, [2] via a secondary source.

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